Cross Reality Workflow - Final Presentation

Matthias Janßen - 1871808

Agenda

- Initial Workflow
- Changed Workflow
- Cross Reality
- Extended Workflow
- Why XR
- Future Improvements
- Demo

Initial Workflow

Workflow Description

- Planning of mail routes
  - Packages and Letters
  - Need to be sorted into correct mail van
  - Based on Address

- Problems
  - Operator must know all addresses and routes
  - Look up on maps can be difficult
  - Potentially needs to know number of packages (no route for one package)
  - Address might not even exist

Changed Workflow

Changed Workflow

- Final Loading step
- Less technologically advanced areas
  - no conveyor belts
  - no automatic loading
- All packages in one room
- need to be moved into actual vans directly

Cross Reality

Cross Reality - Improvements

- Allow user to switch into VR
  - Map of city
  - Routes marked
  - Current scanned address highlighted
  - Overview of all packages added to each route

- Invalid address check

- Actual loading process stays unchanged

Cross Reality - Type of System

- Transitional CR System
  - Primary Type
  - Core of the system
  - User switches from VST to full VR

- Substitutional
  - Substitutional elements
  - Scanner still visible in VR

Extended Workflow

Extended Workflow

- “Pile” of packages
- Operator picks a package and scans label
- Address is shown
- Operator adds package to a route
- If unsure:
  - Operator switches into VR
  - Address is highlighted to make decision easier

Why XR

Why XR

- Operator needs to move around the room freely
- Don’t want to move to a desk for every unknown address

Future Improvements

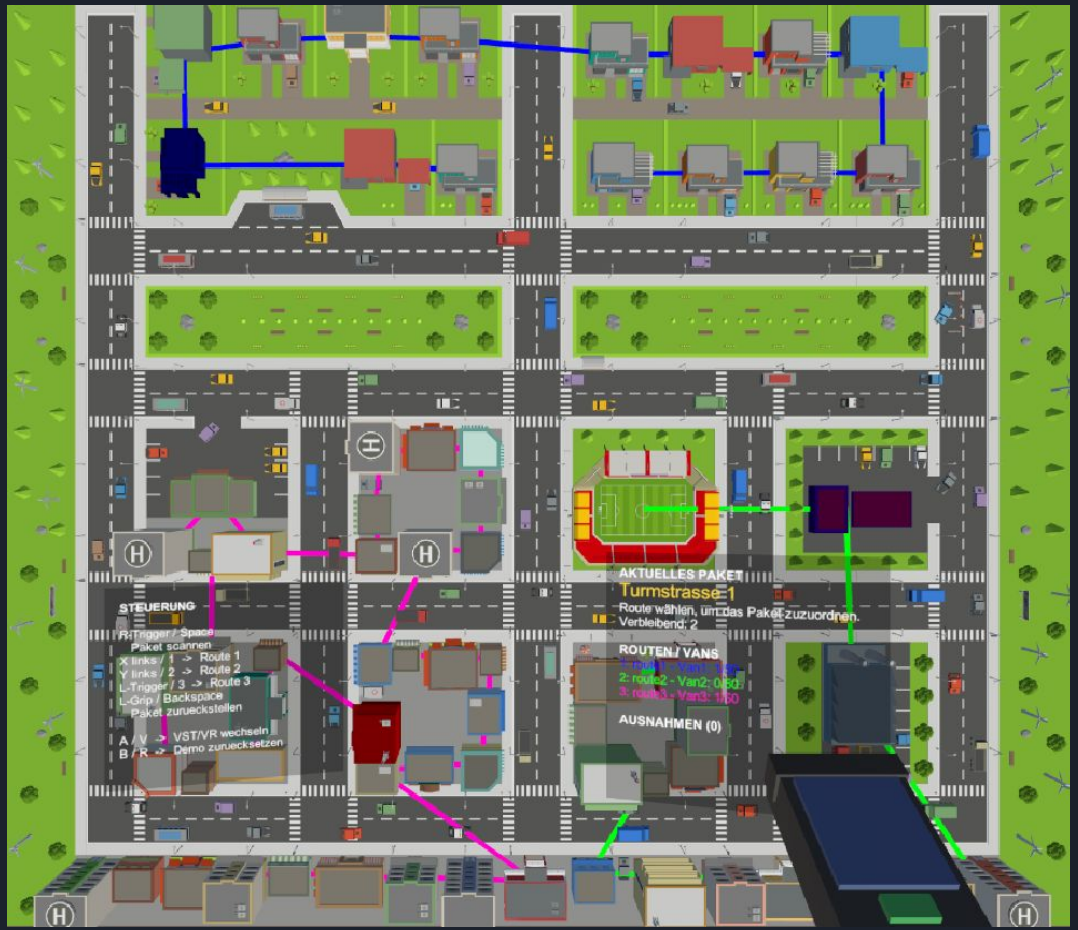
Future Improvements

- Collaboration
- Tracking of packages and automatic address reading
- Tracking which van a package is moved to (no input device needed)
- With real scanning device:
  - left controller not needed anymore, route selection on the scanner

Demo

Demo

- VR Map
- All addresses belonging to a route are connected
- Current address highlighted in red
- Addresses with packages highlighted in corresponding colour
- Van capacity shown



Questions?